

EXPLICIT STABILITY THRESHOLDS OF BLOW-UPS OF PROJECTIVE SPACE ALONG LINEAR SUBSPACES

ABSTRACT. We compute, in closed form, the stability threshold (Fujita–Odaka δ -invariant) of two classical families of Fano manifolds: the blow-up $X_{n,m}$ of $\mathbb{P}^n$ along a linear subspace $\mathbb{P}^m$ with $0 \leq m \leq n-2$, and the blow-up $Y_{a,b}$ of $\mathbb{P}^n$ along a disjoint union of two complementary linear subspaces $\mathbb{P}^a \sqcup \mathbb{P}^b$ with $a+b = n-1$. The key input is an elementary but exact evaluation of the barycenter of the anticanonical polytope, which converts the Blum–Jonsson toric formula into the closed expression

$$\delta(X_{n,m}) = \frac{(n+1)F_{n,m}}{(n+1)F_{n,m} + (k-1)^k(m+2)^{m+1}}, \quad F_{n,m} = \int_{k-1}^{n+1} t^{k-1}(n+1-t)^m dt,$$

where $k = n-m$, together with an analogous formula for $Y_{a,b}$. In both families the threshold is computed by the valuation of an exceptional divisor, and $\delta(Y_{a,b}) = 1$ if and only if $a = b$. For $m = 0$ our formula recovers the value of Zhang–Zhou for $\mathbb{P}^1$ -bundles, and for $m = n-2$ it shows that an upper bound of Benammar Ammar–Massonnet–Yin for projective bundles over curves is attained at the anticanonical polarization. We determine the exact asymptotic behavior of $\delta(X_{n,m})$ in the three natural regimes; in particular $\delta(\text{Bl}_{\text{pt}}\mathbb{P}^n) \rightarrow \tanh(1)$ as $n \rightarrow \infty$, and $\delta(X_{n,m}) \sim \sqrt{\pi/(2\theta(1-\theta)n)}$ when $m \sim \theta n$, so the family contains smooth Fano manifolds with stability threshold decaying to zero at an explicit rate. As $\delta < 1$ throughout the first family, we obtain the exact greatest Ricci lower bounds $R(X_{n,m}) = \delta(X_{n,m})$, extending computations of Székelyhidi, Li, and Zhang–Zhou, and quantifying the classical K-instability of these manifolds observed by Matsushima and Futaki.

1. INTRODUCTION

By the solution of the Yau–Tian–Donaldson conjecture, a Fano manifold admits a Kähler–Einstein metric if and only if it is K-polystable; see [CDS15, Tia15] for the smooth case and [BBJ21, LXZ22] and the references therein for the general theory. The *stability threshold*, or δ -invariant, introduced by Fujita–Odaka [FO18] and developed by Blum–Jonsson [BJ20], detects K-stability numerically: a Fano variety X is K-semistable if and only if $\delta(X) \geq 1$, and uniformly K-stable if and only if $\delta(X) > 1$. The invariant has become the fundamental quantitative tool in the explicit K-stability program, from the resolution of the Calabi problem for Fano threefolds [ACC⁺23] and the Abban–Zhuang method [AZ22], to the ongoing construction of explicit K-moduli spaces of Fano threefolds [ACD⁺25, EJP25, KLPZ26]. For K-unstable Fano manifolds the invariant retains strong differential-geometric meaning: by Berman–Boucksom–Jonsson and Cheltsov–Rubinstein–Zhang [BBJ21, CRZ19], the greatest Ricci lower bound of Tian [Tia92, Szé11] satisfies

$$(1) \quad R(X) = \min\{1, \delta(X)\},$$

and by Blum–Liu–Zhou [BLZ22] a divisorial valuation computing $\delta(X) < 1$ induces an *optimal destabilization* of X with uniquely determined twisted K-polystable degeneration.

Despite this, the list of Fano manifolds for which δ is known *exactly* in all dimensions is short. Zhang–Zhou [ZZ21] computed δ for $\mathbb{P}^1$ -bundles $\mathbb{P}_V(L^{-1} \oplus \mathcal{O}_V)$ with L proportional to $-K_V$; Zhuang [Zhu20] treated products; Benammar Ammar–Massonnet–Yin [BMY24]

recently studied arbitrary polarizations on projective bundles over curves; and Hwang–Kim–Park [HKP23] computed the greatest Ricci lower bounds of horospherical manifolds of Picard number one, exhibiting a family with $R \rightarrow 0$.

In this paper we add to this list two classical families of toric Fano manifolds that have played a distinguished role in the history of canonical metrics, and we record the (perhaps surprising) fact that their stability thresholds admit clean closed formulas in all dimensions.

The two families. For $0 \leq m \leq n - 2$ let

$$X_{n,m} = \text{Bl}_{\mathbb{P}^m} \mathbb{P}^n$$

be the blow-up of $\mathbb{P}^n = \mathbb{P}_{\mathbb{C}}^n$ along an m -dimensional linear subspace, a smooth toric Fano manifold of Picard rank 2; we write $k = n - m \geq 2$ for the codimension of the center. Via projection from $\mathbb{P}^m$, the manifold $X_{n,m}$ is the projectivization of $\mathcal{O}_{\mathbb{P}^{k-1}}(1) \oplus \mathcal{O}_{\mathbb{P}^{k-1}}^{\oplus(m+1)}$, so this family also realizes all higher-rank analogues of the $\mathbb{P}^1$ -bundles of [ZZ21] over projective space. The automorphism group of $X_{n,m}$ is non-reductive, so $X_{n,m}$ carries no Kähler–Einstein metric by Matsushima [Mat57]; the case $X_{2,0} = \text{Bl}_{\text{pt}} \mathbb{P}^2$ is the original example, and Székelyhidi [Szé11] famously computed $R(X_{2,0}) = 6/7$.

For $a, b \geq 1$ with $a + b = n - 1$ let

$$Y_{a,b} = \text{Bl}_{\mathbb{P}^a \sqcup \mathbb{P}^b} \mathbb{P}^n$$

be the blow-up of $\mathbb{P}^n$ along two disjoint (necessarily complementary) linear subspaces of dimensions a and b . This is a smooth toric Fano manifold of Picard rank 3, isomorphic to a $\mathbb{P}^1$ -bundle over $\mathbb{P}^a \times \mathbb{P}^b$. The manifold $Y_{1,2} = \text{Bl}_{\mathbb{P}^1 \sqcup \mathbb{P}^2} \mathbb{P}^4$ is Futaki’s original example of a Fano manifold with reductive automorphism group and non-vanishing Futaki invariant [Fut83]; the family was recently revisited by Delcroix [Del22] (see also [DH21]), who showed that $Y_{a,b}$ carries coupled Kähler–Einstein metrics for suitable (a, b) , and that $Y_{a,b}$ is K-unstable whenever $a \neq b$.

Both families are toric, so K-(semi)stability is decided by the barycenter of the anticanonical polytope [WZ04, BB13, Ber16], and, crucially for us, δ is given by the formula of Blum–Jonsson [BJ20, §7] recalled in Section 2. The point of this paper is that for these two families the relevant barycenter integrals collapse, thanks to an exact primitive (Lemma 2.2), to explicit rational numbers in closed form.

Main results. Throughout, $k = n - m$ and

$$(2) \quad F_{n,m} = \int_{k-1}^{n+1} t^{k-1} (n+1-t)^m dt = \frac{(k-1)! m!}{n!} \sum_{j=0}^{k-1} \binom{n}{j} (k-1)^j (m+2)^{n-j} \in \mathbb{Q}_{>0}.$$

Theorem A. *Let $0 \leq m \leq n - 2$ and $k = n - m$. Then*

$$\delta(X_{n,m}) = \frac{(n+1) F_{n,m}}{(n+1) F_{n,m} + (k-1)^k (m+2)^{m+1}} < 1.$$

Moreover $\delta(X_{n,m})$ is computed by ord_E , where $E \subset X_{n,m}$ is the exceptional divisor of the blow-up, and E is the unique torus-invariant prime divisor computing it.

Equivalently, in probabilistic form: if $B \sim \text{Bin}(n, \frac{k-1}{n+1})$, then $\delta(X_{n,m}) = (1 + \beta)^{-1}$ with

$$(3) \quad \beta = (m+1) \cdot \frac{k-1}{n+1} \cdot \frac{\mathbf{P}(B = k-1)}{\mathbf{P}(B \leq k-1)}.$$

For $m = 0$, Theorem A recovers the value $\delta = ((n+1)^{n+1} - (n+1)(n-1)^n) / ((n+1)^{n+1} + (n-1)^{n+1})$ of Zhang–Zhou [ZZ21, Theorem 1.1] for the $\mathbb{P}^1$ -bundle $\text{Bl}_{\text{pt}}\mathbb{P}^n = \mathbb{P}_{\mathbb{P}^{n-1}}(\mathcal{O}(-1) \oplus \mathcal{O})$, and for $n = 2$ Székelyhidi’s 6/7. For $m = n-2$ (codimension-two center) the formula collapses to the strikingly simple

$$(4) \quad \delta(X_{n,n-2}) = \frac{2n+2}{3n+1},$$

which coincides with the upper bound $s_1 = \frac{n(n+1)(a\mu+b)}{a((r+1)(a\mu_{\max}+b)+(n-r)(a\mu_{\min}+b))}$ of Benammar Ammar–Massonnet–Yin [BMY24, Theorem 1.7] for $X_{n,n-2} = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}(1) \oplus \mathcal{O}^{\oplus(n-1)})$ at the anticanonical polarization: their upper bound is therefore attained in this family (their exact formula is proved only for polarizations close to the base).

Theorem B. *Let $a, b \geq 1$ with $a + b = n - 1$, and set*

$$F_{a,b} = \int_b^{b+2} t^b (n+1-t)^a dt, \quad D_{a,b} = b^{b+1}(a+2)^{a+1} - (b+2)^{b+1}a^{a+1}.$$

Then

$$\delta(Y_{a,b}) = \left(1 + \frac{|D_{a,b}|}{(n+1)F_{a,b}}\right)^{-1},$$

and $D_{a,b} > 0$ if and only if $a < b$. In particular:

- (i) $\delta(Y_{a,b}) = 1$ if and only if $a = b$, in which case $Y_{a,a}$ is K -polystable and admits a Kähler–Einstein metric;
- (ii) if $a \neq b$ then $Y_{a,b}$ is K -unstable and $\delta(Y_{a,b})$ is computed by ord_E , where E is the exceptional divisor over the smaller-dimensional center.

Theorem B quantifies the K -instability of the family established qualitatively in [Fut83, Del22]; for Futaki’s fourfold we get $\delta(Y_{1,2}) = 125/131$.

The closed formulas make the asymptotic behavior of the thresholds completely transparent. Write $\gamma(s, x) = \int_0^x t^{s-1} e^{-t} dt$ and $\Gamma(s, x) = \int_x^\infty t^{s-1} e^{-t} dt$ for the incomplete gamma functions.

Theorem C. *Fix $\theta \in (0, 1)$.*

- (i) (Fixed center dimension.) *For fixed $m \geq 0$, as $n \rightarrow \infty$,*

$$\delta(X_{n,m}) \longrightarrow \delta_\infty(m) := \frac{e^c \gamma(m+1, c)}{e^c \gamma(m+1, c) + c^{m+1}}, \quad c = m+2.$$

In particular $\delta(\text{Bl}_{\text{pt}}\mathbb{P}^n) \rightarrow \tanh(1) = 0.7615\dots$

- (ii) (Fixed codimension.) *For fixed $k \geq 2$, as $m \rightarrow \infty$,*

$$\delta(X_{n,n-k}) \longrightarrow \frac{H_k}{H_k + k - 1}, \quad H_k = e^{k-1}(k-1)^{1-k} \Gamma(k, k-1),$$

which equals 2/3 for $k = 2$, consistently with (4).

- (iii) (Proportional regime.) *If $m = m(n)$ with $m/n \rightarrow \theta$, then*

$$\sqrt{n} \delta(X_{n,m}) \longrightarrow \sqrt{\frac{\pi}{2\theta(1-\theta)}}.$$

Combining Theorem A with (1) we obtain exact greatest Ricci lower bounds; part (iii) exhibits what is, to our knowledge, the simplest known family of smooth Fano manifolds whose stability thresholds decay to 0, with an explicit rate (compare the odd symplectic Grassmannians of [HKP23]).

Corollary D. *For all $0 \leq m \leq n - 2$ one has $R(X_{n,m}) = \delta(X_{n,m})$, given in closed form by Theorem A. Moreover*

$$\min_{0 \leq m \leq n-2} \delta(X_{n,m}) = \Theta(n^{-1/2}), \quad \text{and} \quad \sqrt{n} \delta(X_{n, \lfloor n/2 \rfloor}) \longrightarrow \sqrt{2\pi}.$$

In particular blow-ups of projective spaces along linear subspaces of roughly half dimension become arbitrarily K-unstable as $n \rightarrow \infty$.

Method and organization. Both families are toric, and by [BJ20, §7] the computation of δ reduces to the barycenter $\mathbf{b}$ of the anticanonical polytope P : one has $\delta = \min_i (1 + \langle \mathbf{b}, v_i \rangle)^{-1}$ over the primitive ray generators v_i of the fan. The barycenter of P is a priori an n -dimensional integral; the two elementary observations that make the paper work are:

- (a) the polytopes of $X_{n,m}$ and $Y_{a,b}$ are obtained from the simplex of $\mathbb{P}^n$ by truncation along *one* linear functional ℓ (respectively a symmetric pair), so a Dirichlet-type pushforward reduces all integrals to one variable (Lemma 2.1); and
- (b) the resulting one-variable integrands admit the exact primitive

$$\frac{d}{dt} \left(-\frac{t^\kappa (W-t)^{W-\kappa}}{W} \right) = t^{\kappa-1} (W-t)^{W-\kappa-1} (t-\kappa), \quad W, \kappa > 0,$$

(Lemma 2.2), which evaluates the barycenter *exactly*, with no series or special functions.

A pleasant feature of the computation is the identity $\langle \mathbf{b}, v \rangle = -\beta/(m+1)$ for *all* rays v of the fan other than those meeting the exceptional data (Proposition 4.1), which pins down the minimizing valuation immediately.

Section 2 collects preliminaries. Section 3 describes the toric geometry of the two families and proves that they are Fano. Section 4 proves Theorems A and B. Section 5 proves Theorem C and Corollary D, together with the corresponding asymptotics for the second family (Proposition 5.1). Section 6 tabulates exact values in low dimensions, identifies the threefold members in the Mori–Mukai classification, and poses two questions. All closed formulas in this paper have been verified independently by exact rational lattice-point counting (Ehrhart interpolation) in dimensions ≤ 4 and against all special values available in the literature.

Acknowledgements. This note is an exercise in applying the toric technology of Blum–Jonsson; its debt to [BJ20, ZZ21, Del22, BMY24] will be obvious to the reader.

2. PRELIMINARIES

2.1. The stability threshold. Let X be a Fano manifold of dimension n (more generally a $\mathbb{Q}$ -Fano variety). For a prime divisor E over X , i.e. a prime divisor on a normal birational model $\pi: Z \rightarrow X$, let $A_X(E) = 1 + \text{ord}_E(K_Z - \pi^*K_X)$ denote the log discrepancy and

$$S_X(E) = \frac{1}{\text{vol}(-K_X)} \int_0^\infty \text{vol}(\pi^*(-K_X) - tE) dt$$

the expected vanishing order. The stability threshold of Fujita–Odaka [FO18] admits the valuative description [BJ20]

$$\delta(X) = \inf_E \frac{A_X(E)}{S_X(E)},$$

the infimum running over all prime divisors over X . Then X is K-semistable iff $\delta(X) \geq 1$ and uniformly K-stable iff $\delta(X) > 1$ [FO18, BJ20]; by [LXZ22] uniform K-stability agrees with K-stability. A valuation ord_E computes $\delta(X)$ if $A_X(E)/S_X(E) = \delta(X)$.

The greatest Ricci lower bound [Tia92, Szé11] is

$$R(X) = \sup\{t \in [0, 1] : \exists \omega \in c_1(X) \text{ Kähler with } \text{Ric } \omega \geq t\omega\},$$

and $R(X) = \min\{1, \delta(X)\}$ by [BBJ21, CRZ19].

2.2. The toric formula of Blum–Jonsson. Let $N \cong \mathbb{Z}^n$ be a lattice, M its dual, and let $X = X_\Sigma$ be a toric $\mathbb{Q}$ -Fano variety given by a complete fan Σ in $N_\mathbb{R}$ with primitive ray generators $v_1, \dots, v_d$. The anticanonical polytope is

$$P = \{x \in M_\mathbb{R} : \langle x, v_i \rangle \geq -1 \text{ for } 1 \leq i \leq d\},$$

and we write $\mathbf{b} = \mathbf{b}(P) \in M_\mathbb{R}$ for its barycenter with respect to Lebesgue measure. Each ray generator v_i defines a torus-invariant prime divisor $D_i \subset X$ with $A_X(\text{ord}_{D_i}) = 1$. By [BJ20, Corollary 7.7 and §7.6],

$$(5) \quad S_X(\text{ord}_{D_i}) = \langle \mathbf{b}, v_i \rangle + 1, \quad \delta(X) = \min_{1 \leq i \leq d} \frac{1}{\langle \mathbf{b}, v_i \rangle + 1},$$

and $\delta(X)$ is computed by one of the valuations ord_{D_i} . Consequently X is K-semistable iff $\mathbf{b} = 0$ [WZ04, BB13, Ber16, BJ20]; if $\mathbf{b} = 0$ then X is in fact K-polystable and admits a Kähler–Einstein metric [WZ04]. For smooth toric X the right-hand side of (5) also equals $R(X)$ when $\mathbf{b} \neq 0$, by [Li11] (see also [Yot22] for a direct combinatorial comparison).

2.3. A Dirichlet-type reduction. We will use the following elementary pushforward computation; we include the proof for completeness.

Lemma 2.1. *Let $p, q \geq 1$ be integers and let $\sigma : \mathbb{R}_{\geq 0}^p \times \mathbb{R}_{\geq 0}^q \rightarrow \mathbb{R}_{\geq 0}^2$ be the map $(y, z) \mapsto (\sum_i y_i, \sum_j z_j) = (s, t)$. The pushforward of Lebesgue measure under σ is*

$$\sigma_*(dy dz) = \frac{s^{p-1}}{(p-1)!} \frac{t^{q-1}}{(q-1)!} ds dt.$$

Moreover, conditionally on (s, t) the coordinates are exchangeable, so that for any region $T \subset \mathbb{R}_{\geq 0}^2$ and $R = \sigma^{-1}(T)$,

$$\frac{\int_R y_i dy dz}{\int_R dy dz} = \frac{1}{p} \cdot \frac{\int_T s \rho(s, t) ds dt}{\int_T \rho(s, t) ds dt}, \quad \rho(s, t) = s^{p-1} t^{q-1},$$

and similarly for the z_j with s replaced by t and p by q . The same statements hold when $p = 0$ (no y -variables), with $\rho = t^{q-1}$.

Proof. The volume of the simplex $\{y \in \mathbb{R}_{\geq 0}^p : \sum y_i \leq s\}$ is $s^p/p!$, so the pushforward of dy under $y \mapsto \sum y_i$ has density $s^{p-1}/(p-1)!$; the product statement follows by Fubini. The second claim follows from the symmetry of dy under permutation of coordinates, which gives $\int_R y_i = \frac{1}{p} \int_R s$. $\square$

2.4. An exact primitive. The next observation is the engine of the paper.

Lemma 2.2. *Let $W > 0$ and $0 < \kappa < W$. Then on $(0, W)$,*

$$\frac{d}{dt} \left(-\frac{t^\kappa (W-t)^{W-\kappa}}{W} \right) = t^{\kappa-1} (W-t)^{W-\kappa-1} (t-\kappa).$$

Consequently, for any $0 \leq \alpha < \beta \leq W$,

$$\int_\alpha^\beta t^{\kappa-1} (W-t)^{W-\kappa-1} (t-\kappa) dt = \frac{\alpha^\kappa (W-\alpha)^{W-\kappa} - \beta^\kappa (W-\beta)^{W-\kappa}}{W}.$$

Proof. Differentiate: the derivative of $-t^\kappa(W-t)^{W-\kappa}/W$ equals $t^{\kappa-1}(W-t)^{W-\kappa-1}[-\kappa(W-t) + (W-\kappa)t]/W = t^{\kappa-1}(W-t)^{W-\kappa-1}(t-\kappa)$, since $-\kappa(W-t) + (W-\kappa)t = W(t-\kappa)$. $\square$

The exponents occurring below always satisfy $(\kappa-1) + (W-\kappa-1) = W-2$, i.e. the two exponents sum to the dimension count forced by $-K$; this is the reason the primitive exists and the barycenter is exactly computable.

3. THE TWO FAMILIES OF FANO MANIFOLDS

We fix the lattice $N = \mathbb{Z}^n$ with standard basis $e_1, \dots, e_n$ and set $e_0 = -(e_1 + \dots + e_n)$, so that the fan with rays $e_0, e_1, \dots, e_n$ (and all cones generated by proper subsets) is the fan of $\mathbb{P}^n$. Blowing up a smooth torus-invariant subvariety corresponds to the star subdivision at the sum of the rays of the corresponding cone [CLS11, §3.3].

3.1. Family one. Let $0 \leq m \leq n-2$ and $k = n-m$. Realize $\mathbb{P}^m \subset \mathbb{P}^n$ as the torus-invariant subspace corresponding to the cone $\langle e_{m+1}, \dots, e_n \rangle$; its blow-up $X_{n,m}$ is the toric variety with rays

$$e_1, \dots, e_n, \quad e_0, \quad u := e_{m+1} + \dots + e_n,$$

where u is the ray of the exceptional divisor E . The anticanonical polytope is

$$(6) \quad P = \left\{ x \in \mathbb{R}^n : x_i \geq -1 \ (1 \leq i \leq n), \quad \sum_{i=1}^n x_i \leq 1, \quad \sum_{j=m+1}^n x_j \geq -1 \right\}.$$

Write H for the pullback of the hyperplane class and E for the exceptional divisor, so $-K_{X_{n,m}} = (n+1)H - (k-1)E$.

Proposition 3.1. $X_{n,m}$ is Fano for every $0 \leq m \leq n-2$.

Proof. The linear system $|H-E|$ is the pullback of the hyperplanes through $\mathbb{P}^m$; since blowing up resolves the projection $\mathbb{P}^n \dashrightarrow \mathbb{P}^{k-1}$ from $\mathbb{P}^m$, it is base-point free, hence nef, and H is nef. Now

$$-K_{X_{n,m}} = (k-1)(H-E) + (m+2)H.$$

By the toric Kleiman criterion [CLS11, Theorem 6.3.13] it suffices to check positivity on torus-invariant curves C . If $H \cdot C > 0$ then $-K \cdot C \geq (m+2)H \cdot C > 0$. If $H \cdot C = 0$ then C is contracted by the blow-down $X_{n,m} \rightarrow \mathbb{P}^n$, hence lies in a fiber $\mathbb{P}^{k-1}$ of $E = \mathbb{P}^m \times \mathbb{P}^{k-1} \rightarrow \mathbb{P}^m$, where $C \equiv \lambda \ell$ for a line ℓ with $E \cdot \ell = -1$ and $\lambda > 0$; then $-K \cdot C = (k-1)\lambda > 0$. $\square$

Remark 3.2. Projection from $\mathbb{P}^m$ exhibits $X_{n,m} \cong \mathbb{P}_{\mathbb{P}^{k-1}}(\mathcal{O}(1) \oplus \mathcal{O}^{\oplus(m+1)})$, a $\mathbb{P}^{m+1}$ -bundle over $\mathbb{P}^{k-1}$. For $m=0$ this is the $\mathbb{P}^1$ -bundle case covered by [ZZ21]; for $m \geq 1$ the bundle has fiber dimension ≥ 2 and lies outside the scope of the formulas of [ZZ21, BMY24].

3.2. Family two. Let $a, b \geq 1$ with $a+b = n-1$. Realize the two centers as the torus-invariant subspaces $L_a = \mathbb{P}^a$ corresponding to $\langle e_{a+1}, \dots, e_n \rangle$ and $L_b = \mathbb{P}^b$ corresponding to $\langle e_0, e_1, \dots, e_a \rangle$; they are disjoint and complementary. The blow-up $Y_{a,b}$ is the toric variety with rays

$$e_1, \dots, e_n, \quad e_0, \quad u_1 := e_{a+1} + \dots + e_n, \quad u_2 := e_0 + e_1 + \dots + e_a = -u_1,$$

with u_1, u_2 the rays of the exceptional divisors E_1 (over L_a) and E_2 (over L_b). The anticanonical polytope is

$$(7) \quad Q = \left\{ x \in \mathbb{R}^n : x_i \geq -1 \ (1 \leq i \leq n), \quad \sum_i x_i \leq 1, \quad -1 \leq \sum_{j=a+1}^n x_j \leq 1 \right\}.$$

Since $\text{codim } L_a = b+1$ and $\text{codim } L_b = a+1$, we have $-K_{Y_{a,b}} = (n+1)H - bE_1 - aE_2$.

Proposition 3.3. $Y_{a,b}$ is Fano for all $a, b \geq 1$ with $a + b = n - 1$.

Proof. As in Proposition 3.1, $|H - E_1|$ and $|H - E_2|$ are base-point free (they resolve the projections from L_a, L_b), and

$$-K_{Y_{a,b}} = b(H - E_1) + a(H - E_2) + 2H.$$

For a torus-invariant curve C with $H \cdot C > 0$ we get $-K \cdot C \geq 2H \cdot C > 0$. If $H \cdot C = 0$ then C lies in a fiber of $E_1 \rightarrow L_a$ or of $E_2 \rightarrow L_b$, and as before $-K \cdot C = b\lambda > 0$ or $a\lambda > 0$. $\square$

Remark 3.4. Projecting from each center exhibits $Y_{a,b}$ as the $\mathbb{P}^1$ -bundle $\mathbb{P}_{\mathbb{P}^a \times \mathbb{P}^b}(\mathcal{O}(1, 0) \oplus \mathcal{O}(0, 1))$; see [Del22, §2]. Since $\mathcal{O}(1, 0) \otimes \mathcal{O}(0, 1)^{-1} = \mathcal{O}(1, -1)$ is not proportional to $K_{\mathbb{P}^a \times \mathbb{P}^b}$, this family, too, is outside the scope of [ZZ21]; and $Y_{a,b}$ is not a product, so [Zhu20] does not apply. Note $Y_{a,b} \cong Y_{b,a}$.

4. PROOFS OF THEOREMS A AND B

4.1. Family one: barycenter. Throughout this subsection fix $0 \leq m \leq n - 2$, put $k = n - m \geq 2$, $W = n + 1$, and let P be the polytope (6) with barycenter $\mathbf{b}$. Consider the linear functional $\ell(x) = \langle x, u \rangle = \sum_{j>m} x_j$ and put

$$\beta := \langle \mathbf{b}, u \rangle.$$

By the $\mathfrak{S}_m \times \mathfrak{S}_k$ -symmetry of P we may write $\mathbf{b} = (A, \dots, A, C, \dots, C)$ with A repeated m times and C repeated k times, so $\beta = kC$.

Substituting $w = x + (1, \dots, 1)$ maps P onto

$$P^+ = \left\{ w \in \mathbb{R}_{\geq 0}^n : \sum_i w_i \leq W, \sum_{j>m} w_j \geq k - 1 \right\},$$

and applying Lemma 2.1 with $p = m$, $q = k$, $s = \sum_{i \leq m} w_i$, $t = \sum_{j>m} w_j$, all barycenter computations happen on the triangle

$$T = \{(s, t) : s \geq 0, k - 1 \leq t, s + t \leq W\} \quad \text{with density} \quad \rho = s^{m-1} t^{k-1}.$$

Let $\mathbb{E}[\cdot]$ denote the mean with respect to $\rho ds dt$ on T . Integrating out s (for $m \geq 1$) gives the marginal density $t^{k-1}(W - t)^m/m$ on $[k - 1, W]$, hence

$$(8) \quad \mathbb{E}[t] = \frac{G_{n,m}}{F_{n,m}}, \quad F_{n,m} = \int_{k-1}^W t^{k-1}(W - t)^m dt, \quad G_{n,m} = \int_{k-1}^W t^k(W - t)^m dt,$$

which also holds verbatim when $m = 0$. By Lemma 2.1, $C = \mathbb{E}[t]/k - 1$ and, when $m \geq 1$, $A = \mathbb{E}[s]/m - 1$; since conditionally on t the variable s has density s^{m-1} on $[0, W - t]$, we get $\mathbb{E}[s | t] = \frac{m}{m+1}(W - t)$ and hence $\mathbb{E}[s] = \frac{m}{m+1}(W - \mathbb{E}[t])$.

Proposition 4.1. With $\beta = kC = \mathbb{E}[t] - k$ we have

$$\langle \mathbf{b}, e_j \rangle = \frac{\beta}{k} \quad (m < j \leq n), \quad \langle \mathbf{b}, e_i \rangle = \langle \mathbf{b}, e_0 \rangle = -\frac{\beta}{m+1} \quad (1 \leq i \leq m), \quad \langle \mathbf{b}, u \rangle = \beta.$$

Proof. The first and last identities are the definitions. For $m \geq 1$,

$$A = \frac{\mathbb{E}[s]}{m} - 1 = \frac{W - \mathbb{E}[t]}{m+1} - 1 = \frac{W - (m+1) - \mathbb{E}[t]}{m+1} = \frac{k - \mathbb{E}[t]}{m+1} = -\frac{\beta}{m+1},$$

using $W = m + k + 1$. Finally $\langle \mathbf{b}, e_0 \rangle = -(mA + kC) = \frac{m\beta}{m+1} - \beta = -\frac{\beta}{m+1}$. For $m = 0$ the middle identities reduce to $\langle \mathbf{b}, e_0 \rangle = -kC = -\beta = -\beta/(m+1)$. $\square$

Lemma 4.2. $\beta > 0$.

Proof. Let $\Delta = \{x : x_i \geq -1, \sum x_i \leq 1\}$ be the polytope of $\mathbb{P}^n$, whose barycenter is 0, and let $\Delta_- = \Delta \cap \{\ell < -1\}$, so that $P = \Delta \setminus \Delta_-$ up to measure zero. Since $\int_{\Delta} \ell dx = 0$,

$$\text{vol}(P) \beta = \int_P \ell dx = - \int_{\Delta_-} \ell dx = \int_{\Delta_-} (-\ell) dx \geq \text{vol}(\Delta_-) > 0,$$

because $-\ell > 1$ on Δ_- , and Δ_- has nonempty interior: the vertex $(-1, \dots, -1)$ of Δ satisfies $\ell = -k \leq -2$. $\square$

Lemma 4.3. $\beta = \frac{(k-1)^k (m+2)^{m+1}}{(n+1) F_{n,m}}$, with $F_{n,m}$ as in (2).

Proof. By (8), $\beta F_{n,m} = G_{n,m} - k F_{n,m} = \int_{k-1}^W t^{k-1} (W-t)^m (t-k) dt$. Apply Lemma 2.2 with $\kappa = k$ and $W = n+1$ (so $W - \kappa = m+1$) on the interval $[k-1, W]$:

$$\int_{k-1}^W t^{k-1} (W-t)^m (t-k) dt = \frac{(k-1)^k (W - (k-1))^{m+1} - 0}{W} = \frac{(k-1)^k (m+2)^{m+1}}{n+1}.$$

For the binomial-sum expression in (2), substitute $t = Wv$: $F_{n,m} = W^n \int_p^1 v^{k-1} (1-v)^m dv$ with $p = \frac{k-1}{n+1}$, and use the standard identity $\int_p^1 v^{k-1} (1-v)^m dv = \frac{(k-1)! m!}{n!} \mathbf{P}(B \leq k-1)$ for $B \sim \text{Bin}(n, p)$, $n = k+m$, i.e. $\mathbf{P}(B \leq k-1) = \sum_{j=0}^{k-1} \binom{n}{j} p^j (1-p)^{n-j}$ with $1-p = \frac{m+2}{n+1}$. $\square$

Proof of Theorem A. By Lemma 4.2 and Proposition 4.1, the pairings of $\mathbf{b}$ with the rays of the fan of $X_{n,m}$ are $-\frac{\beta}{m+1} < 0 < \frac{\beta}{k} < \beta$ (the last inequality is strict because $k \geq 2$), so the maximum is attained precisely at the ray u of the exceptional divisor E . By (5),

$$\delta(X_{n,m}) = \frac{1}{1+\beta} < 1,$$

computed by ord_E and by no other torus-invariant prime divisor, and Lemma 4.3 gives the closed formula

$$\delta(X_{n,m}) = \frac{(n+1) F_{n,m}}{(n+1) F_{n,m} + (k-1)^k (m+2)^{m+1}}.$$

For (3): dividing numerator and denominator of β by $(n+1)^{n+1}$ and using the binomial-sum form of $F_{n,m}$,

$$\beta = \frac{p^k q^{m+1}}{\frac{(k-1)! m!}{n!} \mathbf{P}(B \leq k-1)} = (m+1) p \cdot \frac{\binom{n}{k-1} p^{k-1} q^{m+1}}{\mathbf{P}(B \leq k-1)} = (m+1) p \cdot \frac{\mathbf{P}(B = k-1)}{\mathbf{P}(B \leq k-1)},$$

where $p = \frac{k-1}{n+1}$, $q = 1-p = \frac{m+2}{n+1}$ and $\frac{n!}{(k-1)! m!} = (m+1) \binom{n}{k-1}$.

Finally, for $m = n-2$ (i.e. $k = 2$) we verify (4): integrating by parts,

$$\begin{aligned} F_{n,n-2} &= \int_1^{n+1} t (n+1-t)^{n-2} dt = \left[\frac{-t (n+1-t)^{n-1}}{n-1} \right]_1^{n+1} + \frac{1}{n-1} \int_1^{n+1} (n+1-t)^{n-1} dt \\ &= \frac{n^{n-1}}{n-1} + \frac{n^n}{n(n-1)} = \frac{2n^{n-1}}{n-1}, \end{aligned}$$

whence $\beta = \frac{1^2 \cdot n^{n-1}}{(n+1) F_{n,n-2}} = \frac{n-1}{2(n+1)}$ and $\delta = (1+\beta)^{-1} = \frac{2(n+1)}{2(n+1)+(n-1)} = \frac{2n+2}{3n+1}$. $\square$

4.2. Family two. Fix $a, b \geq 1$, $a + b = n - 1$, $W = n + 1$, and let Q be the polytope (7) with barycenter $\mathbf{b}$. Set $\ell(x) = \langle x, u_1 \rangle = \sum_{j>a} x_j$ and $\beta := \langle \mathbf{b}, u_1 \rangle$. By symmetry $\mathbf{b} = (A, \dots, A, C, \dots, C)$ with A repeated a times, C repeated $b + 1$ times.

After the shift $w = x + (1, \dots, 1)$ and the reduction of Lemma 2.1 with $p = a$, $q = b + 1$, the relevant region and density are

$$T' = \{(s, t) : s \geq 0, b \leq t \leq b + 2, s + t \leq W\}, \quad \rho = s^{a-1} t^b.$$

(The constraint $-1 \leq \ell \leq 1$ becomes $b \leq t \leq b + 2$; note $s + t \leq W$ is implied by $s \leq W - t$.) Exactly as before,

$$\mathbb{E}[t] = \frac{G_{a,b}}{F_{a,b}}, \quad F_{a,b} = \int_b^{b+2} t^b (W - t)^a dt, \quad G_{a,b} = \int_b^{b+2} t^{b+1} (W - t)^a dt,$$

$\beta = (b + 1)C = \mathbb{E}[t] - (b + 1)$, and the identical computation as in Proposition 4.1 (with m replaced by a and k by $b + 1$) yields

$$(9) \quad \langle \mathbf{b}, e_j \rangle = \frac{\beta}{b + 1}, \quad \langle \mathbf{b}, e_i \rangle = \langle \mathbf{b}, e_0 \rangle = -\frac{\beta}{a + 1}, \quad \langle \mathbf{b}, u_1 \rangle = \beta, \quad \langle \mathbf{b}, u_2 \rangle = -\beta.$$

Lemma 4.4. $\beta = \frac{D_{a,b}}{(n + 1)F_{a,b}}$ with $D_{a,b} = b^{b+1}(a + 2)^{a+1} - (b + 2)^{b+1}a^{a+1}$.

Proof. $\beta F_{a,b} = \int_b^{b+2} t^b (W - t)^a (t - (b + 1)) dt$. Apply Lemma 2.2 with $\kappa = b + 1$, $W = n + 1$ (so $W - \kappa = a + 1$), on $[b, b + 2]$:

$$\beta F_{a,b} = \frac{b^{b+1}(W - b)^{a+1} - (b + 2)^{b+1}(W - b - 2)^{a+1}}{W} = \frac{b^{b+1}(a + 2)^{a+1} - (b + 2)^{b+1}a^{a+1}}{n + 1},$$

using $W - b = a + 2$ and $W - b - 2 = a$. $\square$

Lemma 4.5. The function $g(x) = \left(1 + \frac{2}{x}\right)^{x+1}$ is strictly decreasing on $(0, \infty)$. Consequently $D_{a,b} > 0$ iff $a < b$, $D_{a,b} = 0$ iff $a = b$, and $D_{a,b} < 0$ iff $a > b$.

Proof. Dividing by $a^{a+1}b^{b+1} > 0$, the sign of $D_{a,b}$ is that of $g(a) - g(b)$. Set $\phi(x) = \log g(x) = (x + 1) \log\left(1 + \frac{2}{x}\right)$ and substitute $t = 2/x > 0$:

$$\phi'(x) = \log\left(1 + \frac{2}{x}\right) - \frac{2(x + 1)}{x(x + 2)} = \log(1 + t) - \frac{t(2 + t)}{2(1 + t)} =: \psi(t).$$

Then $\psi(0) = 0$ and $\psi'(t) = \frac{1}{1+t} - \frac{2+2t+t^2}{2(1+t)^2} = -\frac{t^2}{2(1+t)^2} < 0$, so $\psi(t) < 0$ for $t > 0$, i.e. $\phi' < 0$. $\square$

Proof of Theorem B. If $a = b$ then $\beta = 0$ by Lemma 4.4 and Lemma 4.5; combined with (9) this gives $\mathbf{b} = 0$, so $Y_{a,a}$ is K-polystable and Kähler–Einstein [WZ04, BB13, Ber16], and $\delta = 1$ by (5).

If $a < b$ then $\beta > 0$ and by (9) the pairings are $-\beta < -\frac{\beta}{a+1} < 0 < \frac{\beta}{b+1} < \beta$, so the unique maximizing ray is u_1 , i.e. the exceptional divisor E_1 over the smaller-dimensional center L_a , and $\delta(Y_{a,b}) = (1 + \beta)^{-1} < 1$. If $a > b$, then by $Y_{a,b} \cong Y_{b,a}$ (or by the same argument with $\beta < 0$ and maximizing ray u_2) the threshold is $(1 + |\beta|)^{-1}$, computed by the exceptional divisor over L_b . Lemma 4.4 gives the stated closed formula in all cases. $\square$

Example 4.6. For Futaki's fourfold $Y_{1,2} = \text{Bl}_{\mathbb{P}^1 \sqcup \mathbb{P}^2} \mathbb{P}^4$: $D_{1,2} = 2^3 \cdot 3^2 - 4^3 \cdot 1 = 8$ and $F_{1,2} = \int_2^4 t^2(5 - t) dt = \frac{100}{3}$, so $\beta = \frac{8}{5 \cdot 100/3} = \frac{6}{125}$ and $\delta(Y_{1,2}) = \frac{125}{131}$.

Remark 4.7. In both families, when $\delta < 1$ the threshold is computed by the valuation of a divisor on X itself. By [BLZ22, Theorem 1.2], ord_E induces an optimal destabilizing test configuration — here, the deformation to the normal cone of E — whose central fiber is a uniquely determined twisted K-polystable Fano variety. It would be interesting to compare this with the optimal degenerations for the H-invariant in the sense of Han–Li.

5. ASYMPTOTICS

We now prove Theorem C. Throughout, $\beta = \beta_{n,m}$ denotes the quantity of Lemma 4.3, so $\delta(X_{n,m}) = (1 + \beta)^{-1}$.

Proof of Theorem C(i). Fix m and let $n \rightarrow \infty$ (so $k = n - m \rightarrow \infty$). Substituting $t = n + 1 - s$ in $F_{n,m}$,

$$\frac{(n+1)F_{n,m}}{(k-1)^k} = \int_0^{m+2} \left(\frac{n+1-s}{k-1}\right)^{k-1} \cdot \frac{n+1}{k-1} s^m ds.$$

For $s \in [0, m+2]$ we have $\left(\frac{n+1-s}{k-1}\right)^{k-1} = \left(1 + \frac{m+2-s}{k-1}\right)^{k-1} \nearrow e^{m+2-s}$ as $k \rightarrow \infty$, and the factor $\frac{n+1}{k-1} \leq m+3$ for $k \geq 2$; by dominated convergence the integral converges to $\int_0^{m+2} e^{m+2-s} s^m ds = e^c \gamma(m+1, c)$ with $c = m+2$. Hence $\beta \rightarrow e^{m+1}/(e^c \gamma(m+1, c))$ and $\delta \rightarrow e^c \gamma(m+1, c)/(e^c \gamma(m+1, c) + e^{m+1})$. For $m = 0$: $\gamma(1, 2) = 1 - e^{-2}$, so $\delta_\infty(0) = \frac{e^2 - 1}{e^2 + 1} = \tanh(1)$. $\square$

Proof of Theorem C(ii). Fix $k \geq 2$ and let $m \rightarrow \infty$. Substituting $t = k - 1 + x$,

$$F_{n,m} = (m+2)^m \int_0^{m+2} (k-1+x)^{k-1} \left(1 - \frac{x}{m+2}\right)^m dx.$$

For $m \geq 2$ the integrand is dominated by $(k-1+x)^{k-1} e^{-x/2} \in L^1(0, \infty)$ and converges pointwise to $(k-1+x)^{k-1} e^{-x}$; hence

$$\int_0^{m+2} (k-1+x)^{k-1} \left(1 - \frac{x}{m+2}\right)^m dx \longrightarrow \int_0^\infty (k-1+x)^{k-1} e^{-x} dx = e^{k-1} \Gamma(k, k-1).$$

Therefore, writing $J_m = \int_0^{m+2} (k-1+x)^{k-1} \left(1 - \frac{x}{m+2}\right)^m dx$ and using $\frac{n+1}{m+2} = \frac{m+k+2}{m+2} \rightarrow 1$,

$$\beta = \frac{(k-1)^k (m+2)^{m+1}}{(n+1)F_{n,m}} = \frac{m+2}{n+1} \cdot \frac{(k-1)^k}{J_m} \longrightarrow \frac{(k-1)^k}{e^{k-1} \Gamma(k, k-1)} = \frac{k-1}{H_k}$$

with $H_k = e^{k-1} (k-1)^{1-k} \Gamma(k, k-1)$. Hence $\delta \rightarrow H_k/(H_k + k - 1)$. For $k = 2$: $\Gamma(2, 1) = 2e^{-1}$, so $H_2 = 2$ and the limit is $2/3$, in agreement with (4). $\square$

Proof of Theorem C(iii). Let $m = m(n)$ with $m/n \rightarrow \theta \in (0, 1)$, and use the probabilistic form (3): $\beta = (m+1)p \mathbf{P}(B = k-1)/\mathbf{P}(B \leq k-1)$ with $B \sim \text{Bin}(n, p)$, $p = p_n = \frac{k-1}{n+1} \rightarrow 1 - \theta$ and $q_n = 1 - p_n \rightarrow \theta$; in particular p_n, q_n are bounded away from 0 and 1.

The point $k-1$ sits essentially at the mean: $k-1 - np_n = \frac{k-1}{n+1} = p_n \in (0, 1)$. By the de Moivre–Laplace local limit theorem (see e.g. [Fel68, VII.3]),

$$\mathbf{P}(B = k-1) = \frac{1 + o(1)}{\sqrt{2\pi n p_n q_n}},$$

and by the central limit theorem $\mathbf{P}(B \leq k-1) \rightarrow \Phi(0) = \frac{1}{2}$. Hence

$$\beta = (1 + o(1)) \cdot \frac{2(m+1)p_n}{\sqrt{2\pi n p_n q_n}} = (1 + o(1)) \cdot \frac{2\theta(1-\theta)n}{\sqrt{2\pi n \theta(1-\theta)}} = (1 + o(1)) \sqrt{\frac{2\theta(1-\theta)n}{\pi}} \rightarrow \infty,$$

so $\delta = (1 + \beta)^{-1} = (1 + o(1)) \beta^{-1} = (1 + o(1)) \sqrt{\frac{\pi}{2\theta(1-\theta)n}}$. $\square$

Proof of Corollary D. The identity $R = \delta$ follows from (1) since $\delta < 1$ by Theorem A, and the limit along $m = \lfloor n/2 \rfloor$ is Theorem C(iii) with $\theta = 1/2$; this also gives the upper bound $\min_m \delta(X_{n,m}) \leq (1 + o(1)) \sqrt{2\pi/n}$.

For the matching lower bound $\min_m \delta(X_{n,m}) \geq c n^{-1/2}$ we bound $\beta = \beta_{n,m}$ from above, uniformly in m , using (3). Write $x = m + 1$, $y = k - 1$ and $u = xy/n$, so $x + y = n$ and $u \leq n/4$. First, since $np = (k - 1) - p$ with $p \in (0, 1)$, the integer $k - 1$ equals $\lceil np \rceil$, and the median of a binomial distribution lies between $\lfloor np \rfloor$ and $\lceil np \rceil$ [KB80]; hence $\mathbf{P}(B \leq k - 1) \geq \frac{1}{2}$ and $\beta \leq 2(m + 1)p \mathbf{P}(B = k - 1)$, with $(m + 1)p = \frac{xy}{n+1} \leq u$. If $u \leq 2$ this already gives $\beta \leq 2u \leq 4$. If $u \geq 2$ and $n \geq 3$, then $npq = \frac{ny(x+1)}{(n+1)^2} \geq \frac{xy}{2n} \cdot \frac{2n^2}{(n+1)^2} \geq \frac{u}{2} \geq 1$, and a standard Stirling estimate gives an absolute constant C_0 with $\max_j \mathbf{P}(B = j) \leq C_0/\sqrt{npq}$ whenever $npq \geq 1$ (see e.g. [Fel68, Ch. VII]); hence

$$\beta \leq 2u \cdot \frac{C_0}{\sqrt{u/2}} = 2C_0\sqrt{2u} \leq C_0\sqrt{2n}.$$

In all cases $\beta \leq C_1\sqrt{n}$ for an absolute constant C_1 , whence $\delta = (1 + \beta)^{-1} \geq c n^{-1/2}$. $\square$

The same techniques give the limit of the second family; we state it for completeness and omit the (identical) dominated-convergence argument.

Proposition 5.1. *Fix $a \geq 1$ and let $b \rightarrow \infty$. Then*

$$\delta(Y_{a,b}) \longrightarrow \frac{C_a}{C_a + e^{-2}(a+2)^{a+1} - a^{a+1}}, \quad C_a = \int_0^2 e^{-s} (a+s)^a ds,$$

a number in $(0, 1)$ by Lemma 4.5. For $a = 1$ the limit is $\frac{2 - 4e^{-2}}{1 + 5e^{-2}} = 0.8699\dots$

6. EXAMPLES, TABLES, AND QUESTIONS

6.1. Low-dimensional values. Table 1 lists the exact thresholds of $X_{n,m}$ for $n \leq 6$, and Table 2 those of $Y_{a,b}$ for $n \leq 7$. For $n = 3$ these are the Fano threefolds $\mathbb{N}^{\circ}2.35$ ($X_{3,0} = \text{Bl}_{\text{pt}}\mathbb{P}^3$), $\mathbb{N}^{\circ}2.33$ ($X_{3,1} = \text{Bl}_{\text{line}}\mathbb{P}^3$) and $\mathbb{N}^{\circ}3.25$ ($Y_{1,1}$, the blow-up of $\mathbb{P}^3$ along two disjoint lines) in the Mori–Mukai classification. The values $\delta(2.35) = \frac{14}{17}$ and $\delta(2.33) = \frac{4}{5}$ refine the K-instability of these families recorded in [ACC⁺23], and $\delta(3.25) = 1$ recovers the Kähler–Einstein property of $\mathbb{N}^{\circ}3.25$. The value $\delta(X_{3,0}) = \frac{14}{17}$ also agrees with the $\mathbb{P}^1$ -bundle formula of [ZZ21], and the toric threefold values agree with the table of [Yot22].

$n \backslash m$	0	1	2	3	4
2	6/7				
3	14/17	4/5			
4	340/421	95/127	10/13		
5	633/793	13/18	12/17	3/4	
6	59514/75139	24661/34901	2506/3721	203/299	14/19

TABLE 1. $\delta(X_{n,m}) = R(X_{n,m})$ for $n \leq 6$. Note the non-monotonicity in m (minimal near $m \approx n/2$), consistent with Theorem C(iii).

n	(a, b)	$\delta(Y_{a,b})$
3	(1, 1)	1 (Kähler–Einstein; №3.25)
4	(1, 2)	125/131 (Futaki’s example)
5	(1, 3)	1797/1927
5	(2, 2)	1 (Kähler–Einstein)
6	(1, 4)	7742/8417
6	(2, 3)	28273/28963
7	(1, 5)	103403/113455
7	(2, 4)	34724/36089
7	(3, 3)	1 (Kähler–Einstein)

TABLE 2. $\delta(Y_{a,b})$ for $n \leq 7$.

6.2. Questions.

Question 6.1. The blow-up of $\mathbb{P}^n$ along a disjoint union of linear subspaces $L_{a_1} \sqcup \cdots \sqcup L_{a_r}$ is again toric whenever the centers are torus-invariant, and Fano in a range of cases. The truncation structure of the polytope persists, but with several functionals ℓ_i ; do the barycenter integrals still evaluate in closed form, e.g. for chains of nested linear centers (Bott towers)?

Question 6.2. By Remark 4.7, each $X_{n,m}$ has an optimal destabilization given by deformation to the normal cone of E . What are the twisted K-polystable central fibers, and how do they compare with the limits of the normalized Kähler–Ricci flow on $X_{n,m}$? For $X_{2,0}$ this is understood by work of Székelyhidi and Zhang–Zhou; the higher-dimensional picture, in particular in the regime $m \sim n/2$ where $\delta \rightarrow 0$, seems new.

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